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BSCS SEC E

Task 1:-

Type and save the following programs in Visual Studio. Run these programs and observe their output.

```
#include<iostream>
using namespace std;
int main()
{
  char letter;
   letter = 'A';
   cout << letter << '\n';
   letter = 'B';
   cout << letter << '\n';
   cin >> letter;
   return 0;
}
```

Out put :-

A

B

2)

```
#include <iostream>

using namespace std;

int main ()

{

char letter;
 letter = 65;
 cout << letter << '\n';
  letter = 66;
  cout << letter << '\n';
   cin >> letter;
  return 0;

}
```

Output :-

A

B.

3)

```
#include <iostream>
using namespace std;
int main;
{
int letter;
letter = 'A';
cout << letter << '\n';
letter: 'B';
cout <<<letter << '\n';
cin >> letter;
return 0;
}
```

Output:-

A

B

Task 3:-

Write the output of following code.

Task 2:-

```
# include <iostream>
using namespace std;
int main ()
{
  int val11 , val12 , val13;
  cout << "Please enter your 3 numbers: ";
  cin >> val11 >>val12 >> val13;
  cout << " \n" << val11 << end 1;
cout << val12 <<end 1;
cout << val13 << end1;
  return 0 ;
}
```

Output:-

10

20

30

```
    break;
    case '/': (b != 0)
    cout << "Result: " << (a / b) << endl;
    else
     cout << "Division by zero error!" << endl;
     break;
    }
    return 0;
```

Output:-

Enter first number: 12

Enter Second number: 8

Enter operator (+ - * /): +

Result: 20

Task 4:- Write C++ program to calculate the distance between the two points. Note: x1, y1, x2, y2 are all double values.

formula: $d = (x_2 - x_1)^2 + (y_2 - y_1)^2$.

```
#include <iostream>
#include <cmath>
using namespace std;
 int main () {
  int x1, y1, x2, y2;
  double d;
```

Task 2:-

Two integers as input from the user apply athematic operations on them (+, -, *, /) as print them on screen.

```
#include <iostream>
using namespace std;
int main()
{
    int a, b;
    char op;
    cout << "Enter first number: ";
    cin >> a;
    cout << "Enter second number: ";
    cin >> b;
    cout << "Enter operator (+, -, *, /): ";
    cin >> op;
case '+':
    cout << "Result: " << (a+b) << endl;
    break;
case '-':
    cout << "Result: " << (a-b) << endl;
    break;
case '*':
    cout << "Result: " << (a*b) << endl;
```

```
cout << "Enter value of x1: ";
cin >> x1;
cout << "Enter value of y1 : ";
cin >> y1;
cout << "Enter value of x2 : ";
cin >> x2;
Cout << "Enter value of y2 : ";
cin >> y2;
d = sqrt ( pow(x2 - x1), 2) + pow(y2-y1), 2);
cout << "Distance between the two points is : "<<d;
return d;
}
```

Output:

Distance between the two points is : 5

Task 5:- 1) Calculate the Area of Circle (area = PI*r2)

2) Calculate the Area of Rectangle (area = length * width)

3) Calculate the Area of triangle (area = base* height* .5)

```
# include <iostream>
# include <c math>
using namespace std;
int main() {
double r , length , width, base, height, area;
// Circle
```

```
cout <<"Enter radius: ";
cin >> r;
area = M_PI * r * r;
cout << "Area of circle: " <<area<< endl;
// Rectangle
cout << "Enter length: ";
 cin >> length;
 cout << " Enter width: ";
 cin >> width;
area = length * width;
cout <<"Area of rectangle: " << area << endl;
// Triangle
cout << "Enter base:";
cin >>base;
cout <<"Enter height: ";
cin >> height;
area = 0.5 * base * height
cout << " Area of triangle: " <<area<< endl;
return 0; ;
}
```

Output:-

Area of circle : 153.938
Area of rectangle : 60
Area of triangle : 20

Task 6: Total Purchase A coustomer in a store is purchasing five ~~it~~mes. The prices of the five items are:

Price of item 1 = $12.95
Price of item 2 = $ 24.95
Price of item 3 = $ 6 .95
Price of item 4 = $ 14.95
Price of item 5 = $ 3.95

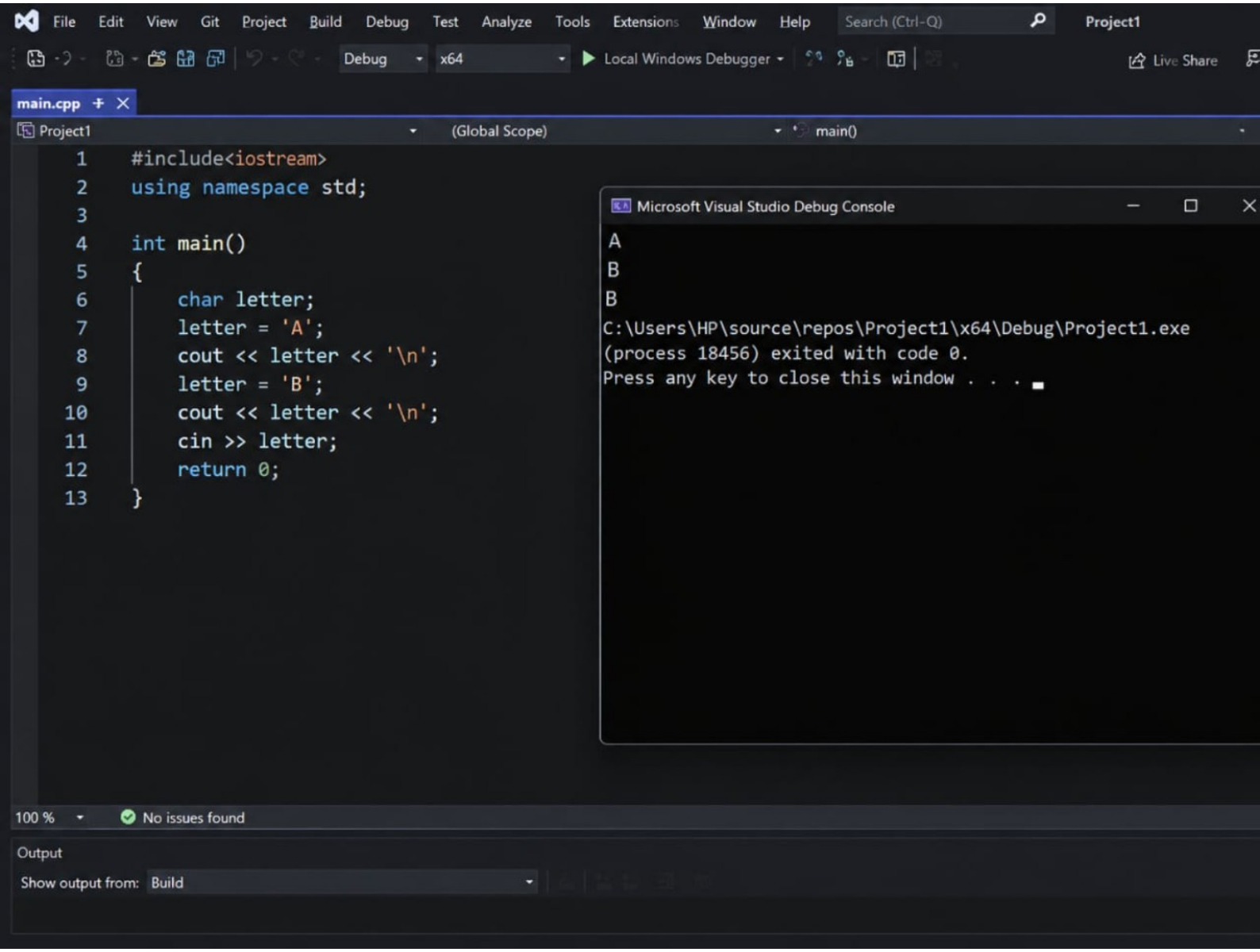
Output :-

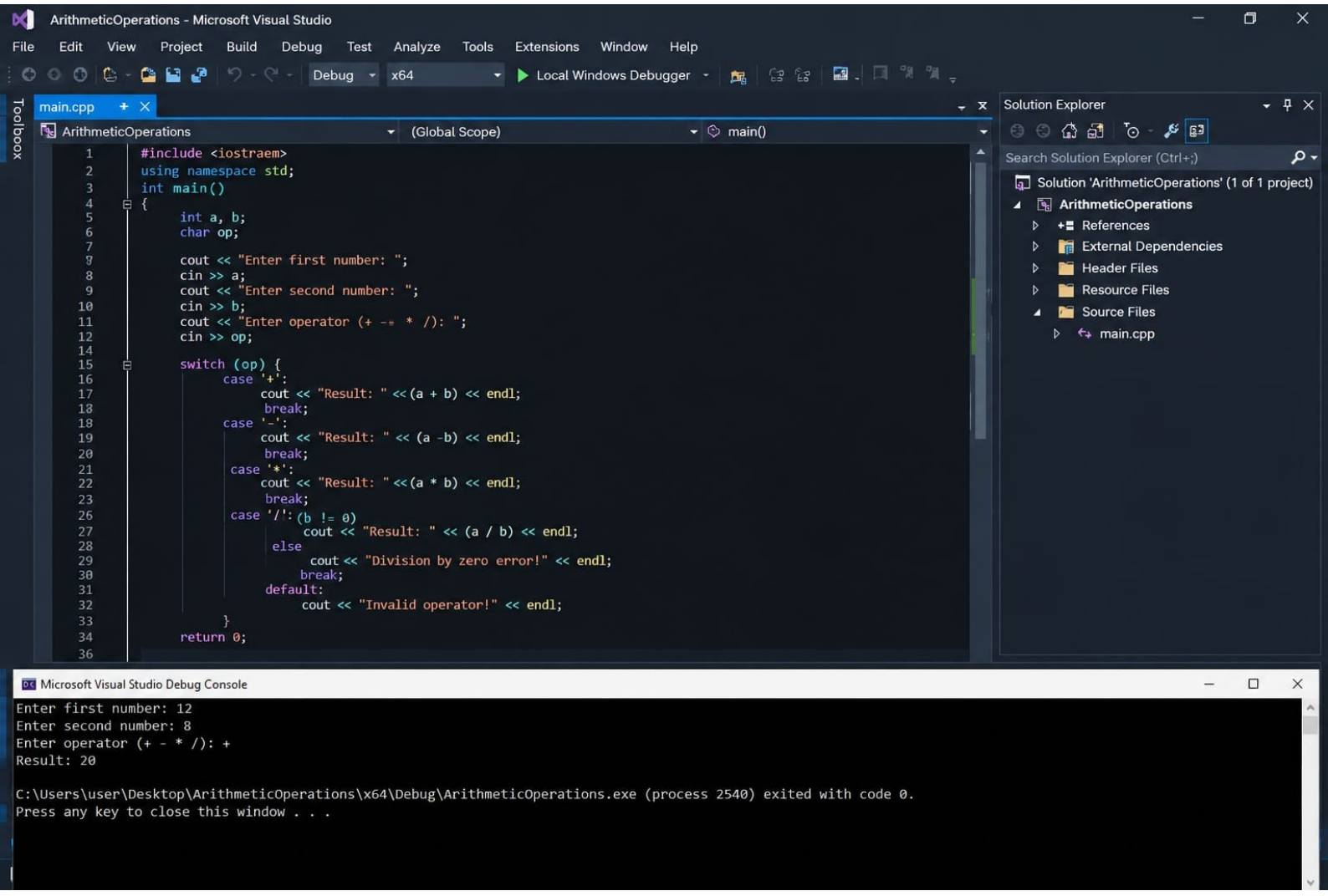
Item 1 Price : $12.95
Item 2 Price : $24.95
Item 3 Price : $6.95
Item 4 Price : $14.95
Item 5 Price : $3.95

Subtotal : $63.75

Sales Tax (6%) : $3.83

Total : $67.58


File Edit View Git Project Build Debug Test Analyze Tools Extensions Window Help
Search (Ctrl-Q)
Project1
Debug
x64
Local Windows Debugger
Live Share
main.cpp
Project1
(Global Scope)
main()
#include<iostream>
using namespace std;
int main()
{
char letter;
letter = 'A';
cout << letter << '\n';
letter = 'B';
cout << letter << '\n';
cin >> letter;
return 0;
}
Microsoft Visual Studio Debug Console
A
B
B
C:\Users\HP\source\repos\Project1\x64\Debug\Project1.exe
(process 18456) exited with code 0.
Press any key to close this window . . .
100 %
No issues found
Output
Show output from: Build


ArithmeticOperations - Microsoft Visual Studio
File Edit View Project Build Debug Test Analyze Tools Extensions Window Help
Debug x64 Local Windows Debugger
main.cpp
ArithmeticOperations (Global Scope) main()
#include <iostraem>
using namespace std;
int main()
{
int a, b;
char op;
cout << "Enter first number: ";
cin >> a;
cout << "Enter second number: ";
cin >> b;
cout << "Enter operator (+ -= * /): ";
cin >> op;
switch (op) {
case '+':
cout << "Result: " <<(a + b) << endl;
break;
case '-':
cout << "Result: " << (a -b) << endl;
break;
case '*':
cout << "Result: " <<(a * b) << endl;
break;
case '/':(b != 0)
cout << "Result: " << (a / b) << endl;
else
cout << "Division by zero error!" << endl;
break;
default:
cout << "Invalid operator!" << endl;
}
return 0;
Solution Explorer
Search Solution Explorer (Ctrl+;)
Solution 'ArithmeticOperations' (1 of 1 project)
ArithmeticOperations
References
External Dependencies
Header Files
Resource Files
Source Files
main.cpp
Microsoft Visual Studio Debug Console
Enter first number: 12
Enter second number: 8
Enter operator (+ - * /): +
Result: 20
C:\Users\user\Desktop\ArithmeticOperations\x64\Debug\ArithmeticOperations.exe (process 2540) exited with code 0.
Press any key to close this window . . .

main.cpp

```cpp
#include <iostream>
using namespace std;
int main()
{
    int val11, val12, val13;

    cout << "Please enter your 3 numbers: ";
    cin >> val11 >> val12 >> val13;

    cout << "\n" << val11 << endl;
    cout << val12 << endl;
    cout << val13 << endl;

    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL powershell

```
PS C:\Users\user\Desktop> cd "C:\Users\user\Desktop" ; if ($?) { g++ main.cpp -o main }
} ; if ($?) { .\main }
Please enter your 3 numbers: 10 20 30
10
20
30

PS C:\Users\user\Desktop>
```

Input:

Please enter your 3 numbers:
10 20 30

Output:

10
20
30

main.cpp - Visual Studio Code

File Edit Selection View Go Run Terminal Help

main.cpp

```cpp
#include <iostream>
#include <cmath>
using namespace std;

int main() {
    int x1, y1, x2, y2;
    double d;
    cout << "Enter value of x1: ";
    cin >> x1;
    cout << "Enter value of y1: ";
    cin >> y1;
    cout << "Enter value of x2: ";
    cin >> x2;
    cout << "Enter value of y2: ";
    cin >> y2;

    d = sqrt( pow((x2 - x1), 2) + pow((y2 - y1), 2) );

    cout << "Distance between the two points is: " << d;

    return 0;
}
```

Microsoft Visual Studio Debug Console

```
Enter value of x1: 1
Enter value of y1: 2
Enter value of x2: 4
Enter value of y2: 6
Distance between the two points is: 5

C:\Users\user\Desktop\programs\Debug\programs.exe
(process 18432) exited with code 0.
Press any key to close this window . . .
```

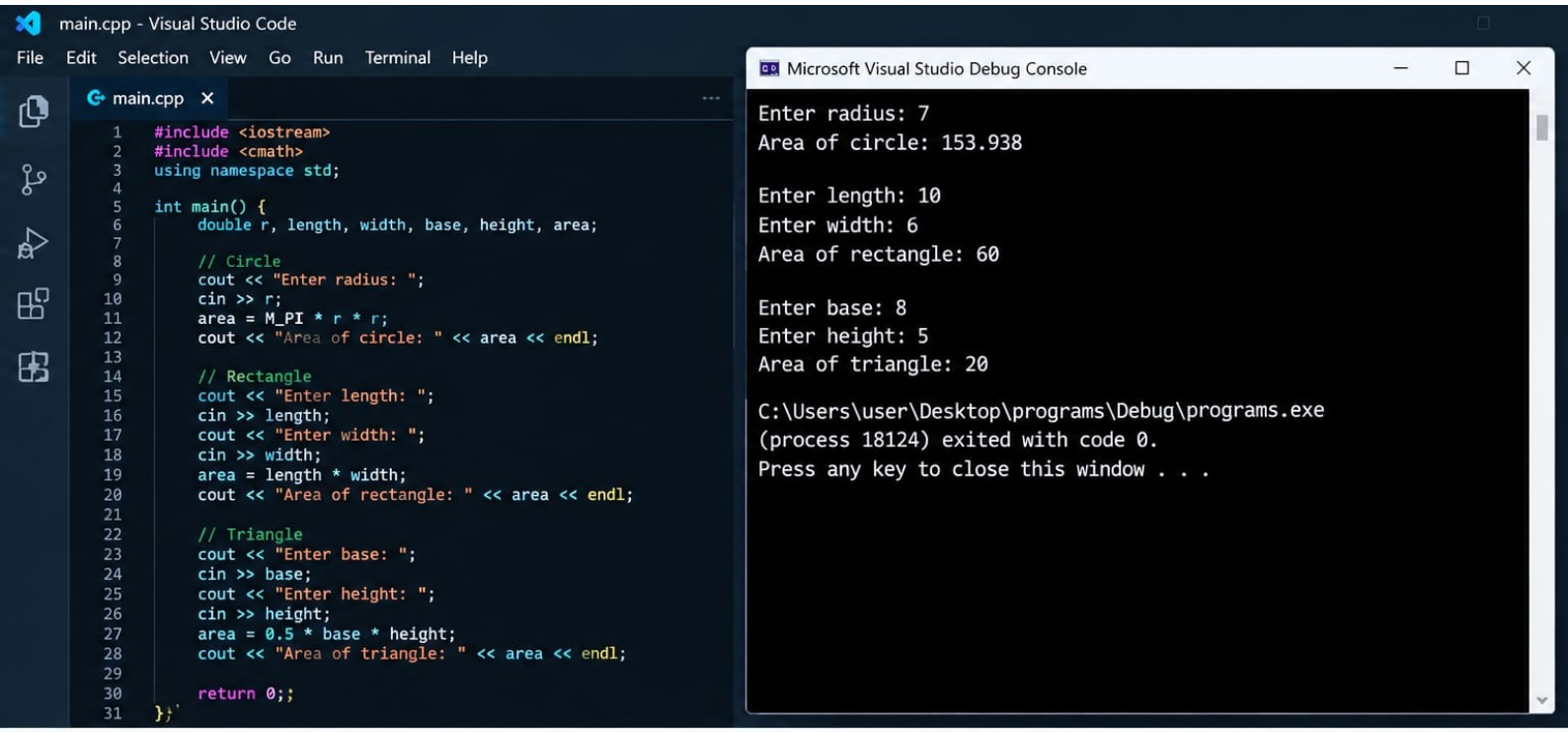
Input:

```
Enter value of x1: 1
Enter value of y1: 2
Enter value of x2: 4
Enter value of y2: 6
```

Output:

```
Distance between the two points is: 5
```

(For reference: $d = \sqrt{((4-1)^2 + (6-2)^2} = \sqrt{3^2 + 4^2} = \sqrt{9+16} = \sqrt{25} = 5$)



Input:

```
Enter radius: 7
Enter length: 10
Enter width: 6
Enter base: 8
Enter height: 5
```

Output:

```
Area of circle: 153.938
Area of rectangle: 60
Area of triangle: 20
```

main.cpp

```cpp
#include <iostream>
using namespace std;

int main() {
    double price1, price2, price3, price4, price5;
    double subtotal, tax, total;

    cout << "Enter price of item 1: ";
    cin >> price1;
    cout << "Enter price of item 2: ";
    cin >> price2;
    cout << "Enter price of item 3: ";
    cin >> price3;
    cout << "Enter price of item 4: ";
    cin >> price4;
    cout << "Enter price of item 5: ";
    cin >> price5;
    subtotal = price1 + price2 + price3 + price4 + price5;
    tax = subtotal * 0.06;
    total = subtotal + tax;

    cout << "\ nSubtotal: $" << subtotal << endl;
    cout << "Sales Tax (6%): $" << tax << endl;
    cout << "Total: $" << total << endl;

    return 0;
}
```

Microsoft Visual Studio Debug Console

```
Enter price of item 1: 12.95
Enter price of item 2: 24.95
Enter price of item 3: 6.95
Enter price of item 4: 14.95
Enter price of item 5: 3.95

Subtotal: $63.75
Sales Tax (6%): $3.83
Total: $67.58

Press any key to close this window . . .
```

Input:

```
Enter price of item 1: 12.95
Enter price of item 2: 24.95
Enter price of item 3: 6.95
Enter price of item 4: 14.95
Enter price of item 5: 3.95
```

Output:

```
Subtotal: $63.75
Sales Tax (6%): $3.83
Total: $67.58
```